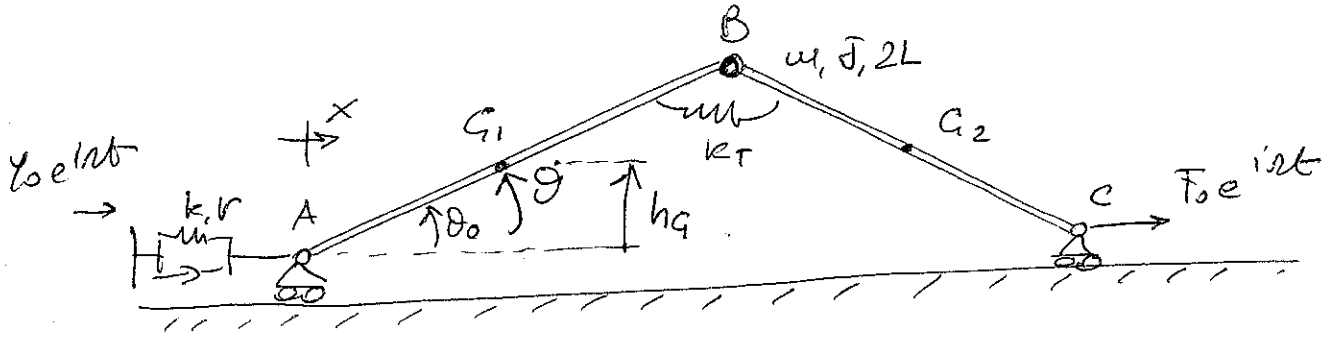




$$x = \begin{Bmatrix} x \\ 0 \\ y \end{Bmatrix}$$

$$E_C = \frac{1}{2} m \dot{x}_{s1}^2 + \frac{1}{2} m \dot{y}_{s1}^2 + \frac{1}{2} J \dot{\theta}^2 + \frac{1}{2} m \dot{x}_{s2}^2 + \frac{1}{2} m \dot{y}_{s2}^2 + \frac{1}{2} J \dot{\theta}^2$$

$$\begin{aligned} X_{g1} &= X + L \cos \theta \\ Y_{g1} &= L \sin \theta \\ X_{g2} &= X + 3L \cos \theta \\ Y_{g2} &= L \sin \theta \end{aligned}$$

$$[M_F] = \text{diag}(m \ m \ J \ m \ m \ J)$$

$$[A_{in}] = \begin{bmatrix} 1 & -L \sin \theta_0 & 0 \\ 0 & L \cos \theta_0 & 0 \\ 0 & 1 & 0 \\ 1 & -3L \sin \theta_0 & 0 \\ 0 & L \cos \theta_0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$V_{el} = \frac{1}{2} k \Delta x^2 + \frac{1}{2} k_T \Delta \theta^2$$

$$\begin{cases} \Delta L_1 = x - y \\ \Delta \theta = 2\theta \end{cases}$$

$$[k_F] = \text{order}(k, k_T)$$

$$[A_k] = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 2 & 0 \end{bmatrix}$$

$$D = \frac{1}{2} r \dot{\alpha}_1^2$$

$$[r_F] = [r] \quad (\Delta r) = [1 \ 0 \ -1]$$

$$V_g = 2mg h_g = 2mg L \sin \theta$$

$$\frac{\partial^2 V_g}{\partial x^2} = \frac{\partial^2 V_g}{\partial x \partial \theta} = 0$$

$$\frac{\partial^2 V_g}{\partial \theta^2} = -2mg L \sin \theta$$

$$\frac{\partial^2 V_g}{\partial x \partial y} = \frac{\partial^2 V_g}{\partial y \partial \theta} = \frac{\partial^2 V_g}{\partial y^2} = 0$$

$$[k_g] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -2mgL \sin \theta_0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

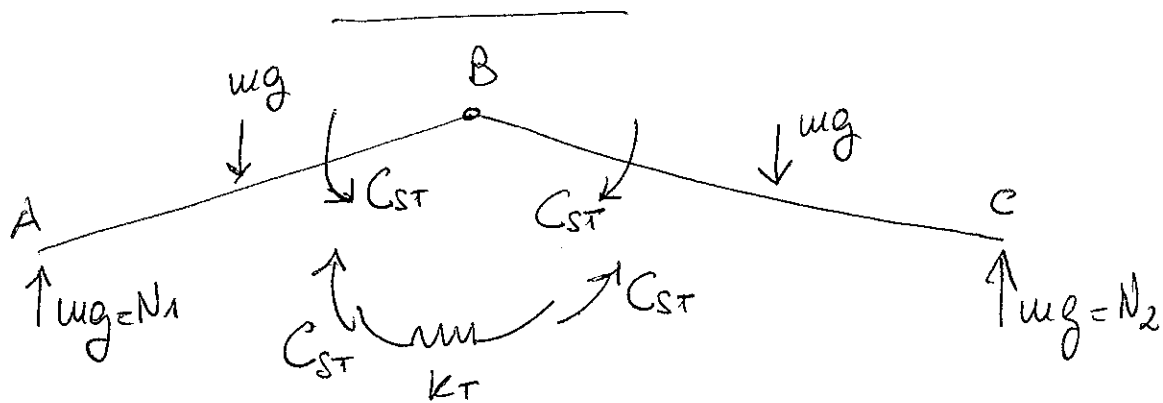
$$\delta L = \vec{F} \times \delta \vec{x}_c =$$

$$x_c = x + 4L \cos \theta$$

$$\delta x_c = \delta x - 4L \sin \theta \delta \theta$$

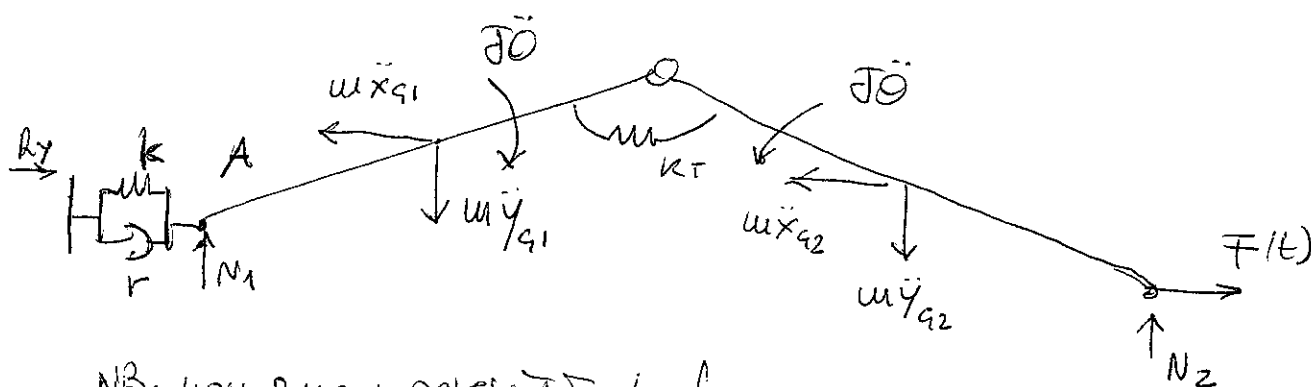
$$\delta L = F \delta x - 4L F \sin \theta_0 \delta \theta$$

$$\rightarrow \underline{Q}_L = \begin{Bmatrix} 1 \\ -4L \sin \theta_0 \end{Bmatrix} F_0 \text{ e inst}$$



$$\sum M_B^{AB} = 0 : C_{ST} + mg L \cos \theta_0 - mg 2L \cos \theta_0 = 0$$

$$C_{ST} = mg L \cos \theta_0$$



NB: non sono rappresentate le forze peso x che determinano la parte statica delle reazioni vincolari N_2 (vedi pag. 2)

$$\begin{cases} \dot{x}_{g1} = \dot{x} - L\dot{\theta}\sin\theta \\ \dot{y}_{g1} = L\dot{\theta}\cos\theta \\ \dot{x}_{g2} = \dot{x} - 3L\dot{\theta}\sin\theta \\ \dot{y}_{g2} = L\dot{\theta}\cos\theta \end{cases} \quad \begin{cases} \ddot{x}_{g1} = \ddot{x} - L\ddot{\theta}\sin\theta - L\dot{\theta}^2\cos\theta \\ \ddot{y}_{g1} = L\ddot{\theta}\cos\theta - L\dot{\theta}^2\sin\theta \\ \ddot{x}_{g2} = \ddot{x} - 3L\ddot{\theta}\sin\theta - 3L\dot{\theta}^2\cos\theta \\ \ddot{y}_{g2} = L\ddot{\theta}\cos\theta - L\dot{\theta}^2\sin\theta \end{cases}$$

$$\begin{cases} \ddot{x}_{g1,LIN} = \ddot{x} - L\ddot{\theta}_0\sin\theta_0 \\ \ddot{y}_{g1,LIN} = L\ddot{\theta}_0\cos\theta_0 \\ \ddot{x}_{g2,LIN} = \ddot{x} - 3L\ddot{\theta}_0\sin\theta_0 \\ \ddot{y}_{g2,LIN} = L\ddot{\theta}_0\cos\theta_0 \end{cases}$$

$$\sum M_A = 0 : N_2 4L \cos\theta_0 + m \ddot{x}_{g1,LIN} L \sin\theta_0 + m \ddot{x}_{g2,LIN} L \sin\theta_0 + \\ - m \ddot{y}_{g1,LIN} L \cos\theta_0 - m \ddot{y}_{g2,LIN} 3L \cos\theta_0 - J_O \ddot{\theta} + J_O \ddot{\theta} = 0$$

$$N_2 = \frac{m \ddot{y}_{g1,LIN} \cos\theta_0 + m \ddot{y}_{g2,LIN} 3 \cos\theta_0 - m \ddot{x}_{g1,LIN} \sin\theta_0 - m \ddot{x}_{g2,LIN} \sin\theta_0}{4 \cos\theta_0}$$